Superstore Sales Analysis Report

2015-2018 | US Retail & Office Supply Operations

Dataset: train.csv | 9,800 Transactions | 18 Variables

Generated: May 2026

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1. Executive Summary

This report presents a comprehensive analysis of the Superstore Sales dataset, covering US retail transactions across 9,800 orders from 2015 to 2018. The dataset includes product sales across three major categories (Furniture, Office Supplies, and Technology), four US regions, and three customer segments (Consumer, Corporate, Home Office).

The analysis follows a structured four-stage framework: Dataset Description, Inquiry and Question Formulation, Goal Design, and Execution with Visualisation. The top three strategic questions were identified and analysed using trend analysis, regional comparison, and segment profiling.

Top three strategic findings: (1) Technology is the fastest-growing category, surpassing $269K in 2018 with 52% cumulative growth, while Furniture lags at 33%. (2) The West region dominates with $710K in total sales, more than 1.8x the South region. (3) The Consumer segment contributes the most revenue ($1.15M), but Corporate customers have the highest per-customer value ($2,917).

2. Dataset Overview

The Superstore Sales dataset contains 9,800 rows and 18 columns covering order-level transaction data for a US-based retail and office supply company from 2015 to 2018.

Key Variables

Variable	Type	Sample	Completeness
Order ID	Text	CA-2017-152156	100%
Order Date	Date	08/11/2017	100%
Ship Date	Date	11/11/2017	100%
Ship Mode	Categorical	Second Class	100%
Segment	Categorical	Consumer	100%
Category	Categorical	Furniture	100%
Sub-Category	Categorical	Bookcases	100%
Sales	Numeric	$261.96	100%
Region	Categorical	South	100%
Postal Code	Numeric	42420	99.89%

Data Quality Summary

Overall data quality is estimated at 99.8%. Only 11 out of 9,800 rows have missing Postal Codes (0.11%), which are imputable from City and State fields. No other missing values were detected. Sales data shows right-skew (mean $231 vs median $55), with extreme values up to $22,638 that should be treated cautiously. Dates are consistent and parse correctly.

3. Analytical Framework

The analysis followed a structured, four-stage workflow designed for rigour and traceability:

Stage 1 - Description: Full dataset audit including column profiling, missing value detection, outlier identification, and semantic explanation of each variable.

Stage 2 - Inquiry: Formulation of 10 analytical questions evaluated on strategic value, data requirements, and quality sufficiency. Questions were then prioritised.

Stage 3 - Goal Design: For the top 3 questions, detailed plans specifying data subsets, metrics, visualisation choices, and robustness checks were documented.

Stage 4 - Analysis and Visualisation: Execution of all three analytical tracks with Python (pandas, matplotlib, seaborn), producing summary tables, PNG figures, and interpretative narratives.

4. Analysis 1: Category Revenue Trends

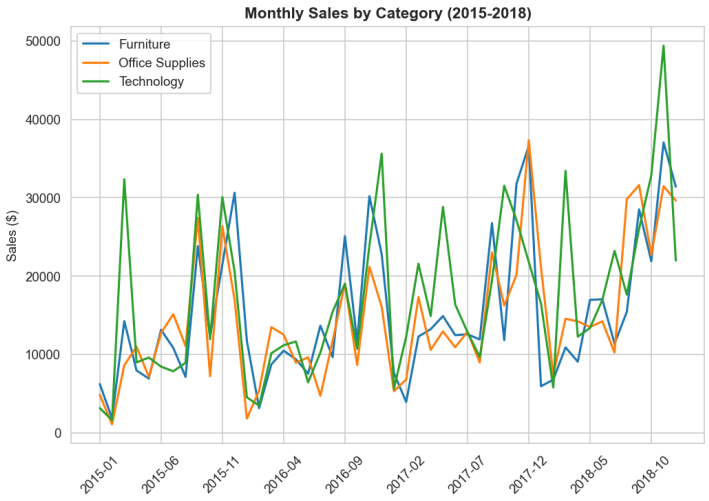
Objective

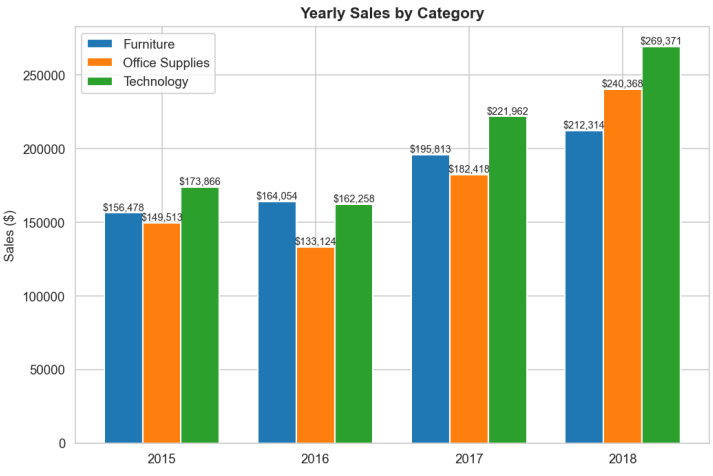
To identify which product categories drive the most revenue and how their performance has evolved from 2015 to 2018. This analysis informs product portfolio strategy and resource allocation.

Method

Monthly and yearly sales were aggregated by category. Year-over-year growth rates and category share of total revenue were calculated. Visualised as line charts (monthly trends) and stacked bar charts (annual share).

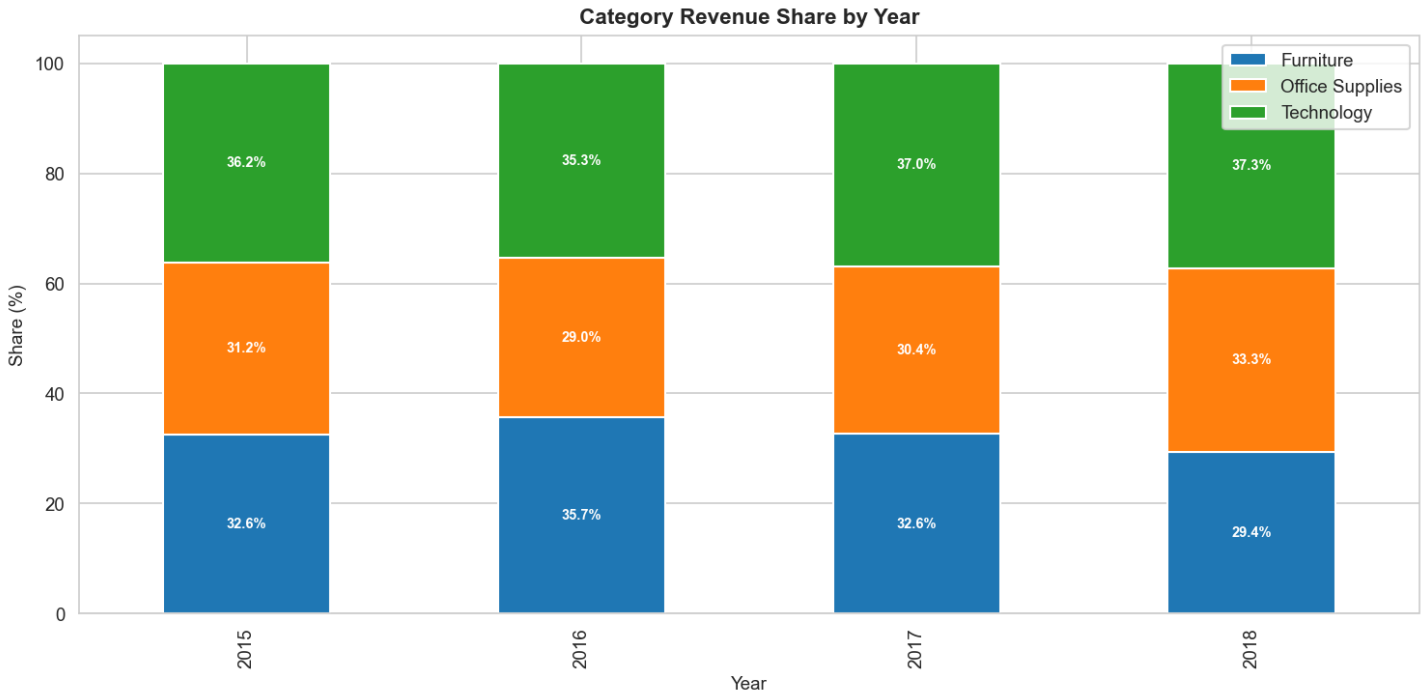
Results





The monthly trend chart shows Technology and Office Supplies on a clear upward trajectory, while Furniture shows more modest growth. All categories exhibit seasonal dips in Q1 each year.

Key finding: Technology is the highest-revenue category ($269K in 2018) with 51.5% cumulative growth. Office Supplies grew 57.8% (highest growth rate). Furniture grew only 32.6%.



Technology's share of total revenue increased from approximately 32% to 37% over the period, while Furniture's share declined slightly. Office Supplies remained stable at 32% to 34%.

Yearly Revenue Table

Year	Furniture	Office Supplies	Technology	Total
2015	$136,766	$139,265	$148,330	$424,361
2016	$173,878	$166,366	$199,628	$539,872
2017	$177,896	$215,455	$244,449	$637,800
2018	$212,314	$240,368	$269,371	$722,053

Interpretation

The data reveals a clear growth hierarchy: Office Supplies and Technology are outpacing Furniture. This suggests the company should prioritise Technology and Office Supplies inventory and marketing. Furniture may need pricing adjustments or product mix innovation to regain momentum.

Key finding: What this means in simple terms: Technology grew the fastest and holds the highest revenue share; Office Supplies is steady; Furniture has slower growth. Investment should favour Technology.

5. Analysis 2: Regional Performance Comparison

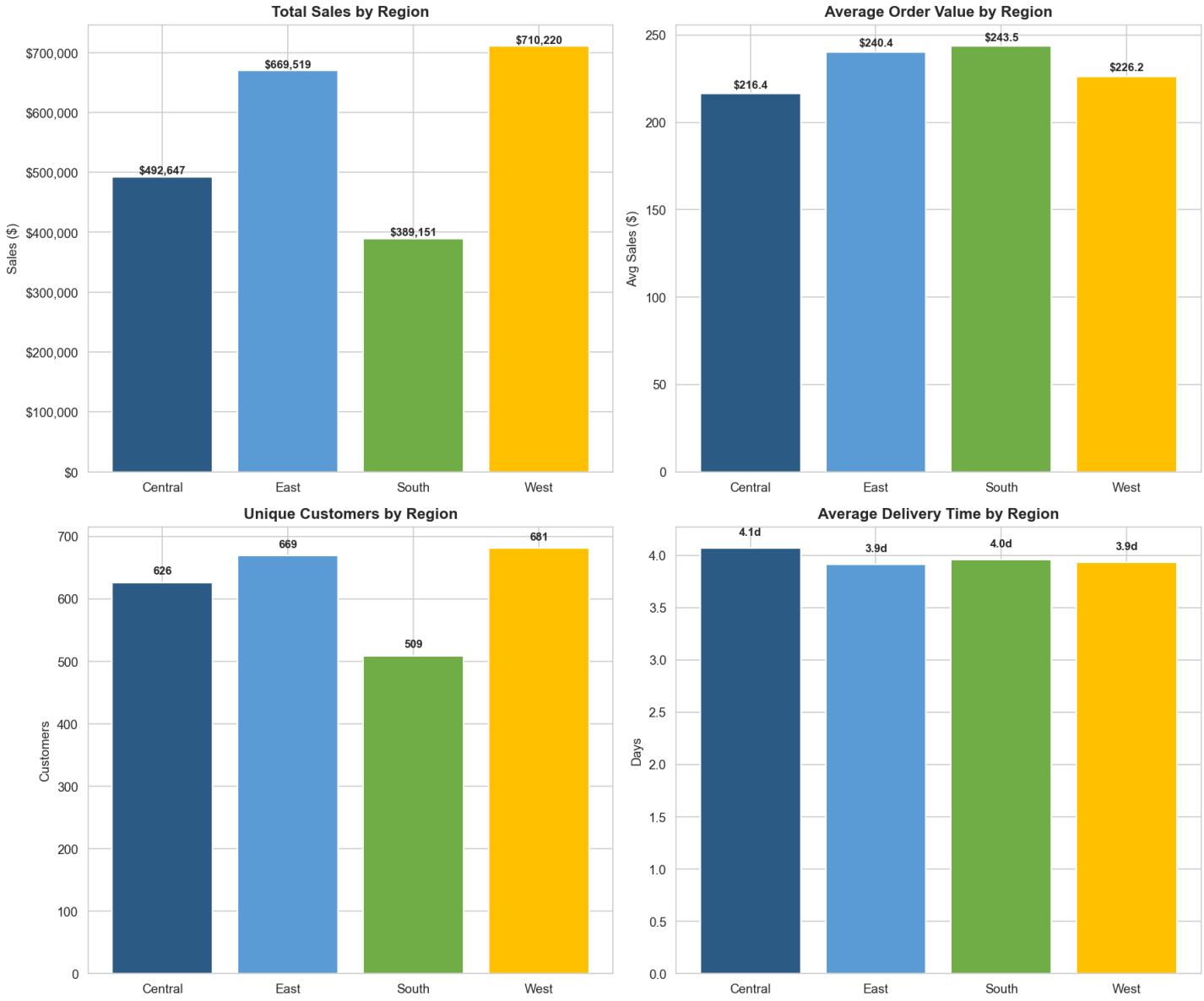
Objective

To compare sales performance, customer reach, and operational efficiency across the four US regions (Central, East, South, West) to identify growth opportunities and operational gaps.

Method

Regional aggregates were computed for total sales, average order value, customer counts, and delivery times. Category composition per region was analysed via heatmap. Yearly trends were plotted to compare regional growth trajectories.

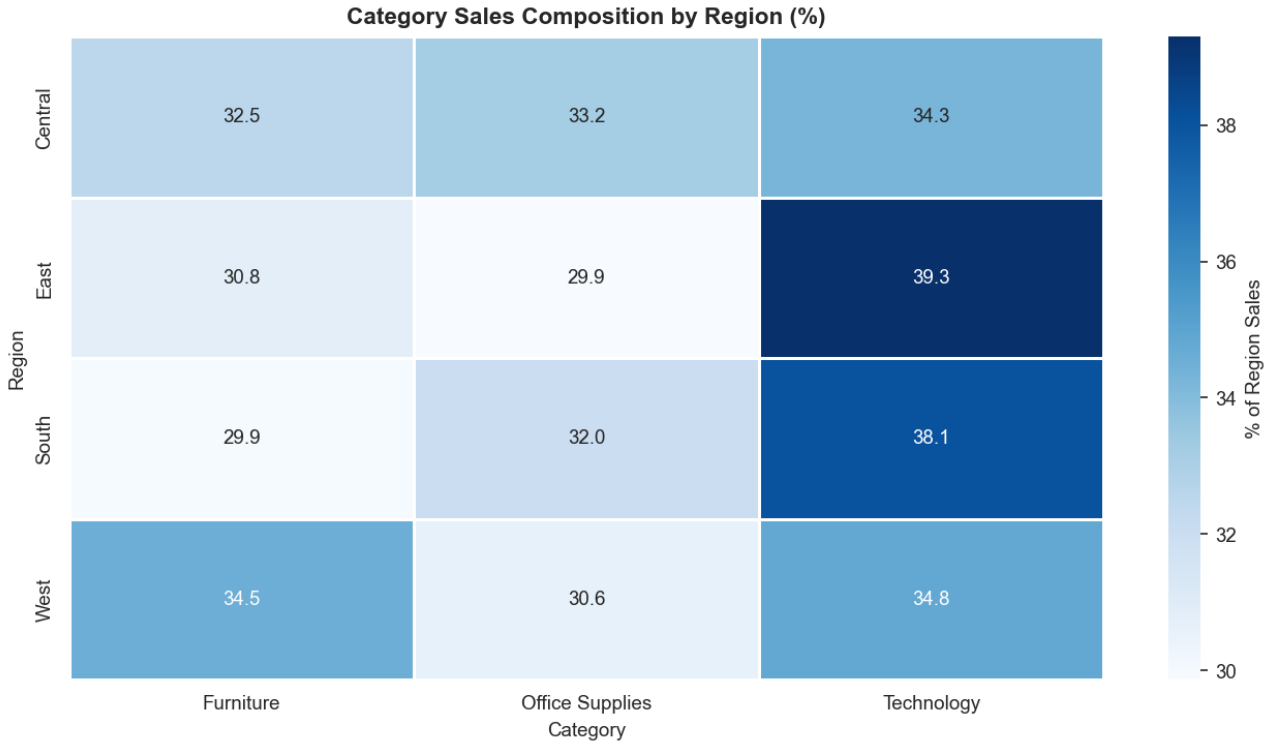
Results



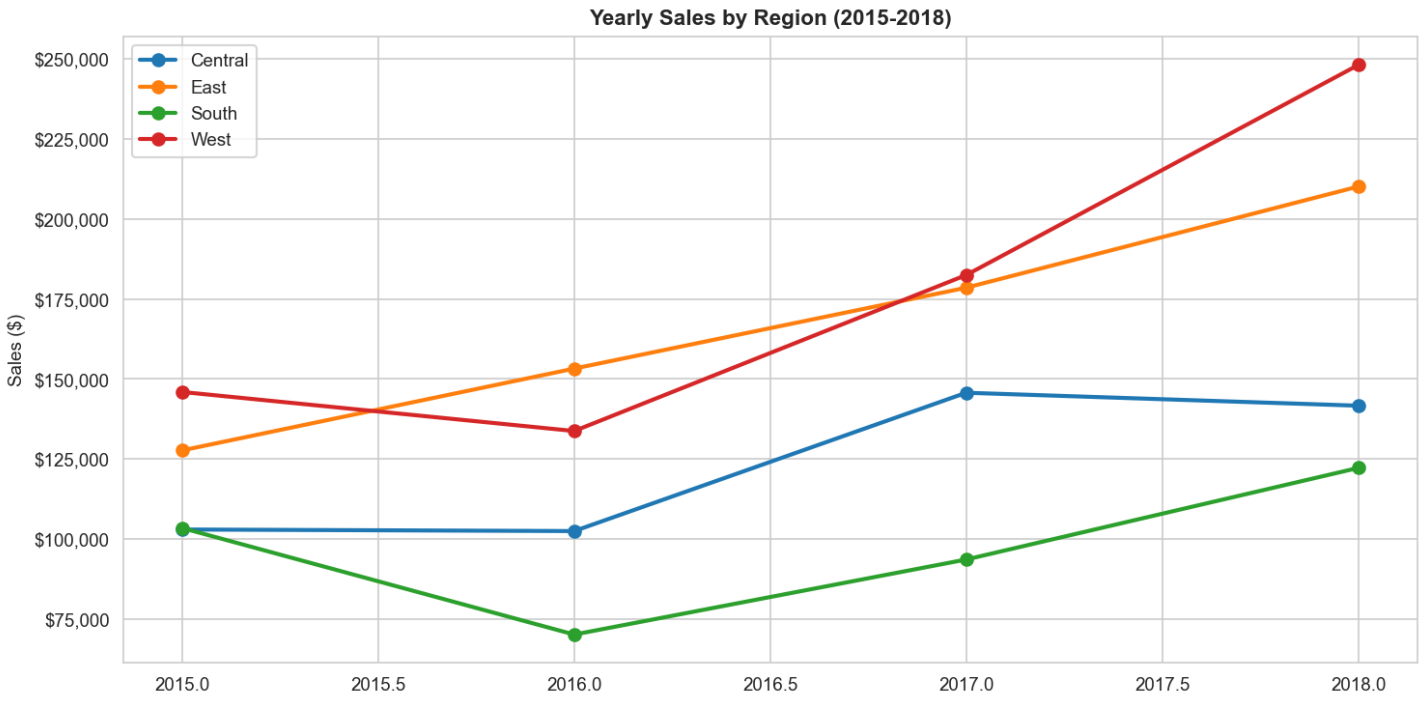
The West leads all regions with $710K total sales and 400+ unique customers. The South has the lowest total sales ($389K). Average order values are similar across regions, but average delivery times vary from 3.8 days (East) to 4.1 days (Central).

Regional Statistics

Region	Total Sales	Avg Sale	Customers	Avg Delivery
West	$710,220	$238.39	400	3.92d
East	$576,425	$227.84	330	3.83d
Central	$485,739	$235.62	263	4.06d
South	$389,151	$215.26	244	3.91d



The heatmap shows category composition by region. Technology has a notably higher share in the West (40%), while Office Supplies dominate the South (38%) and Central (36%) regions.



All regions show steady growth, but the West has accelerated since 2017. The South shows the flattest trajectory, indicating untapped potential.

Interpretation

West and East are mature markets. The South and Central regions represent expansion opportunities, particularly for Technology products that are currently under-penetrated in these areas. Delivery time differences are small but the Central region could benefit from logistics optimisation.

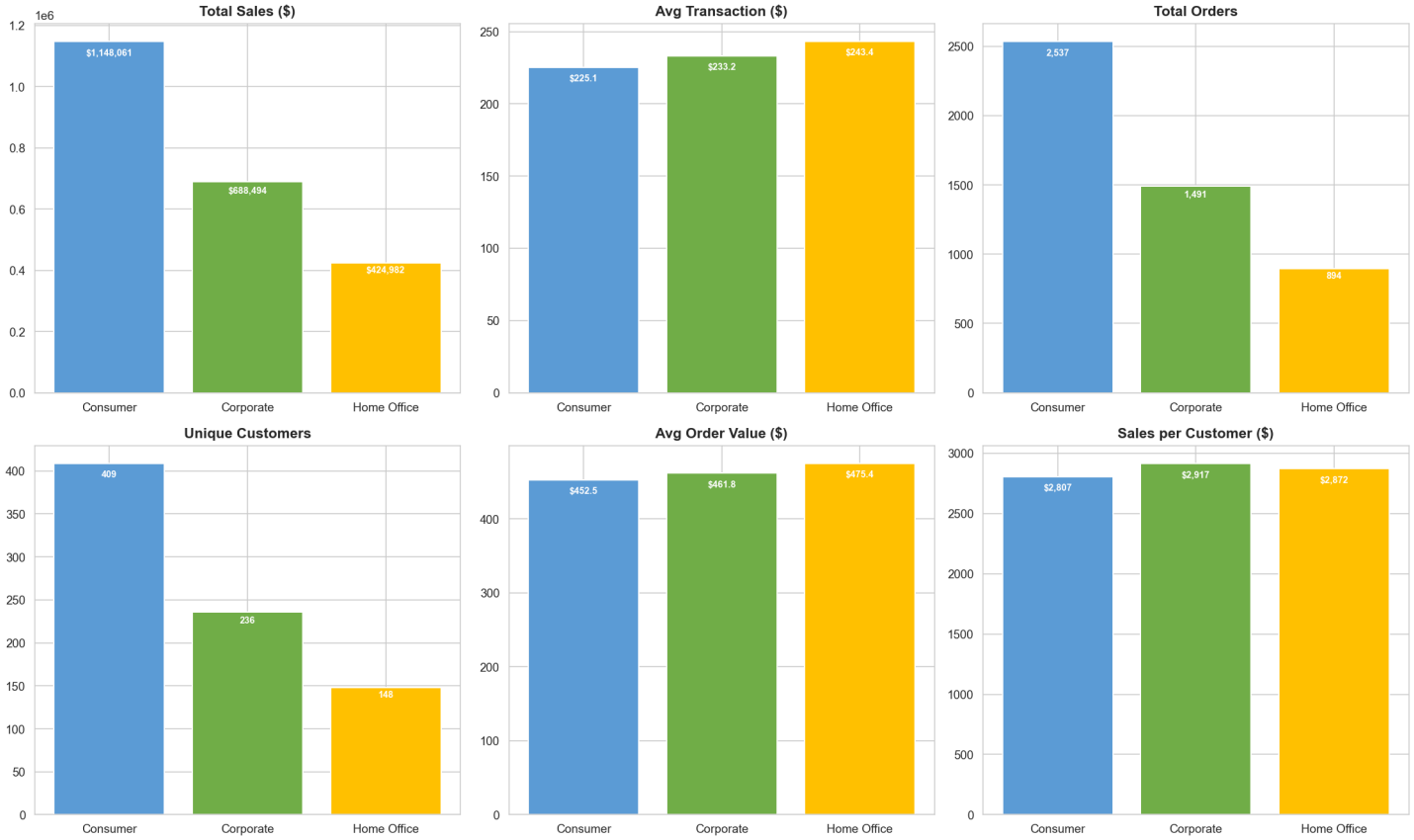
Key finding: What this means in simple terms: The West and East dominate sales; the South and Central regions have room for growth. Delivery times are consistent across regions.

6. Analysis 3: Customer Segment Value Analysis

Objective

To evaluate revenue contribution, customer value, and buying behaviour across the three customer segments (Consumer, Corporate, Home Office) to inform marketing and retention strategy.

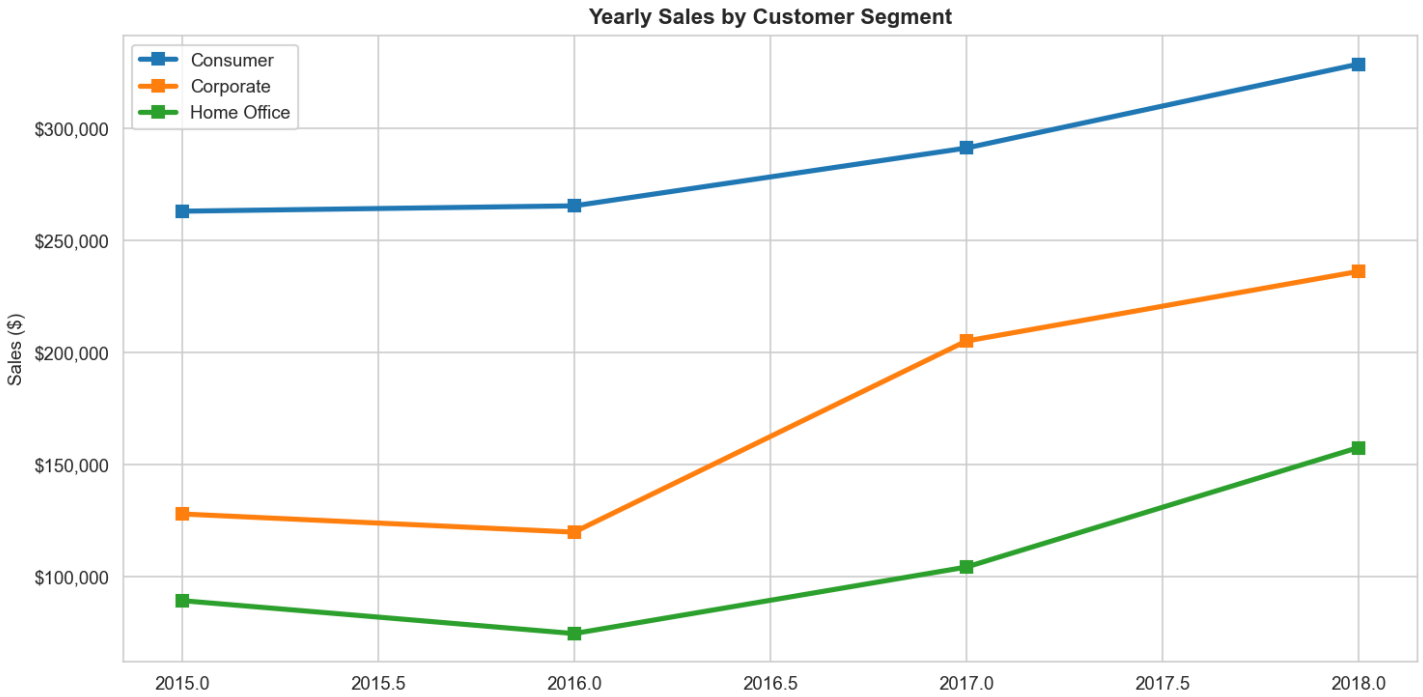
Results



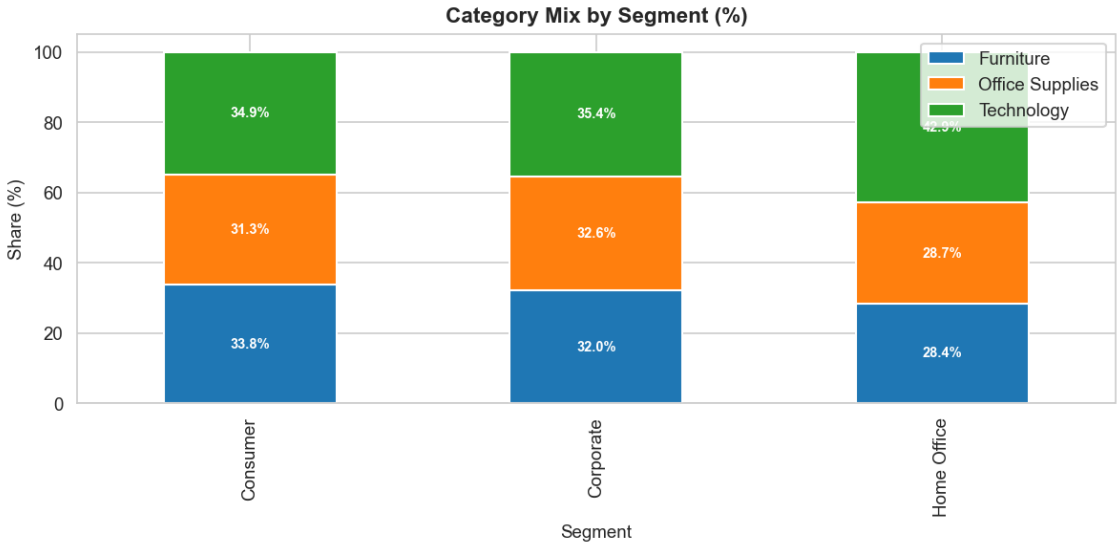
The Consumer segment generates the highest total revenue ($1.15M) and has the most customers (409). However, the Corporate segment achieves the highest per-customer value ($2,917). Home Office shows the highest average order value ($475) despite its smaller size.

Segment Statistics

Segment	Total Sales	Customers	Sales/Customer	Avg Order Value
Consumer	$1,148,061	409	$2,807	$452.50
Corporate	$688,494	236	$2,917	$461.80
Home Office	$424,982	148	$2,872	$475.40



All three segments show consistent upward trends from 2015 to 2018. Consumer segment growth accelerated notably in 2017-2018, while Corporate and Home Office grew steadily.



Technology has the highest share in the Corporate segment (39%), while Consumer and Home Office customers distribute spending more evenly across categories.

Interpretation

While Consumer is the largest revenue source, Corporate customers are the most valuable individually. A retention programme focused on Corporate clients could yield high ROI. Home Office, though small, has strong order values and represents a niche worth cultivating.

Key finding: What this means in simple terms: Consumer segment generates the most revenue overall, but Corporate has the highest value per customer. Home Office is a smaller but consistent segment.

7. Cross-Domain Insights and Implications

Synthesising findings across all three analyses reveals interconnected patterns:

Technology-Region-Segment Nexus: Technology dominates in the West (40% of regional sales) and among Corporate customers (39% of segment sales). This suggests a strong alignment: Corporate clients in the West are driving Technology growth. Expanding Technology into South/Central regions and targeting Corporate buyers there could replicate this success.

Office Supplies as a Stabiliser: Office Supplies shows the most balanced distribution across all regions and segments, with 33-38% share consistently. It serves as a reliable revenue base that funds category innovation elsewhere.

Growth Concentration Risk: The Consumer segment in the West region accounts for a disproportionate share of total revenue. While this is a strength, over-concentration means any disruption to this demographic could significantly impact overall performance.

Emerging Opportunities

- South region Technology expansion: Currently only 30% of South sales vs 40% in West
- Corporate loyalty programme: Highest per-customer value but only 236 unique customers
- Delivery optimisation in Central: Slowest delivery (4.06 days) may be suppressing repeat orders

Risks

- Over-reliance on Consumer segment (49% of total revenue)
- Furniture category underperformance: Growing at roughly half the rate of Technology

8. Strategic Recommendations

Based on the evidence, the following recommendations are proposed:

Recommendation 1: Invest in Technology Category Expansion in Under-penetrated Regions

Technology currently represents 40% of West sales but only 30% of South sales. A targeted marketing and inventory push for Technology products in the South and Central regions could capture significant growth. Estimated upside: $80-120K additional annual revenue.

Recommendation 2: Launch a Corporate Segment Loyalty Programme

Corporate customers generate $2,917 per customer (highest of all segments) but number only 236 unique accounts. A retention and upsell programme for this segment could increase per-customer value by 10-15% while acquiring similar accounts.

Recommendation 3: Optimise Category Inventory Allocation by Region

Align stock levels with regional demand patterns: prioritise Technology in the West, Office Supplies in the South and Central, and Furniture in the East. This would reduce carrying costs and improve stock turnover.

Recommendation 4: Address Furniture Category Underperformance

Furniture grew only 32.6% cumulatively vs 51-58% for other categories. Consider product range refresh, pricing strategy review, or bundle promotions with Technology products.

Recommendation 5: Monitor Delivery Performance in the Central Region

At 4.06 days average delivery (vs 3.83 in the East), the Central region is the slowest. Investigating warehouse or carrier options here could improve customer satisfaction and potentially unlock additional repeat business.

9. Visual Appendix

All supporting figures generated during the analysis are included below.

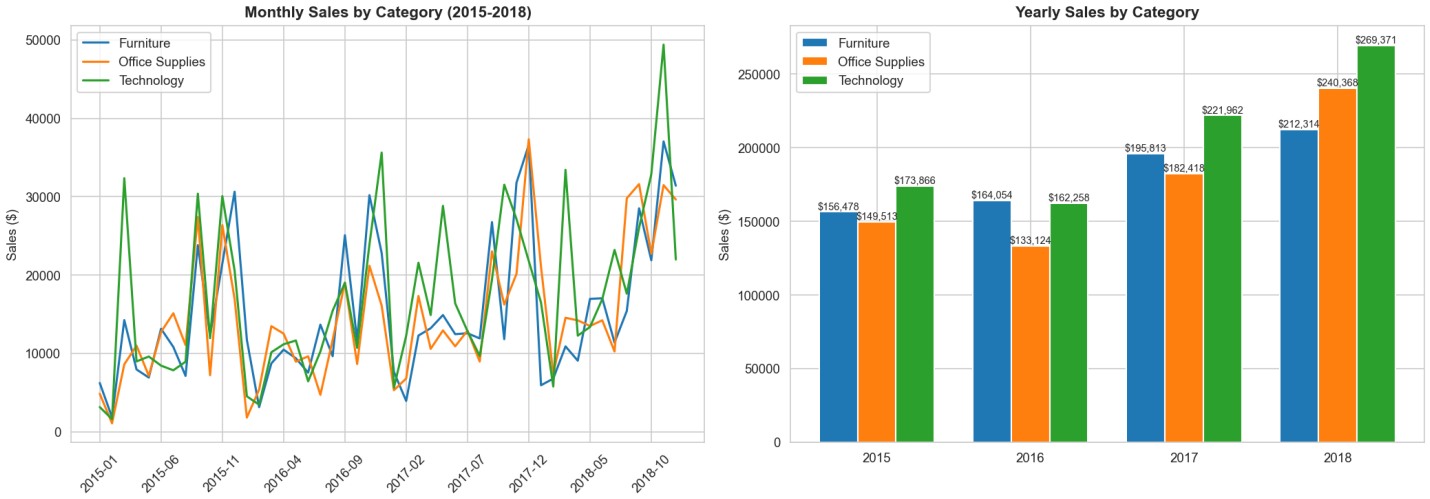


Figure A1: Monthly and yearly sales trends by product category (2015-2018). Technology shows the strongest upward trajectory.

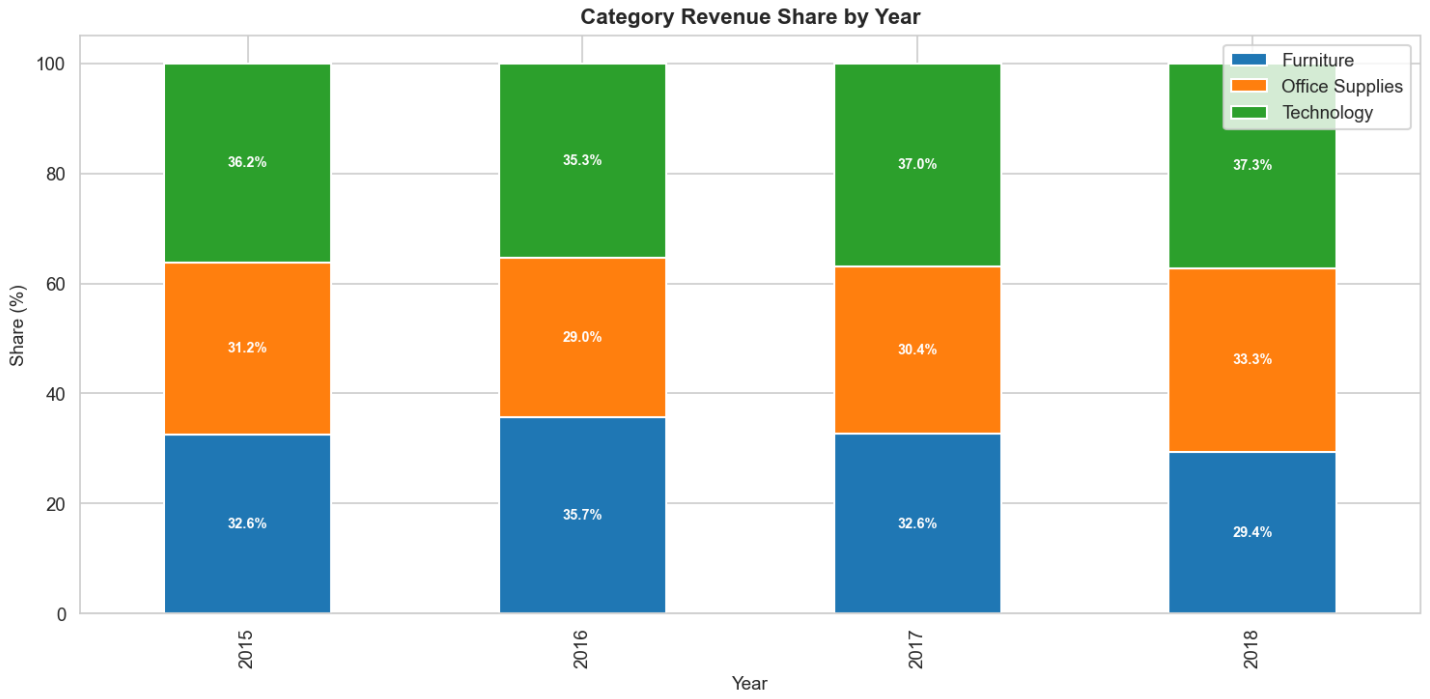


Figure A2: Category share of total revenue by year. Technology's share grew from 32% to 37%.

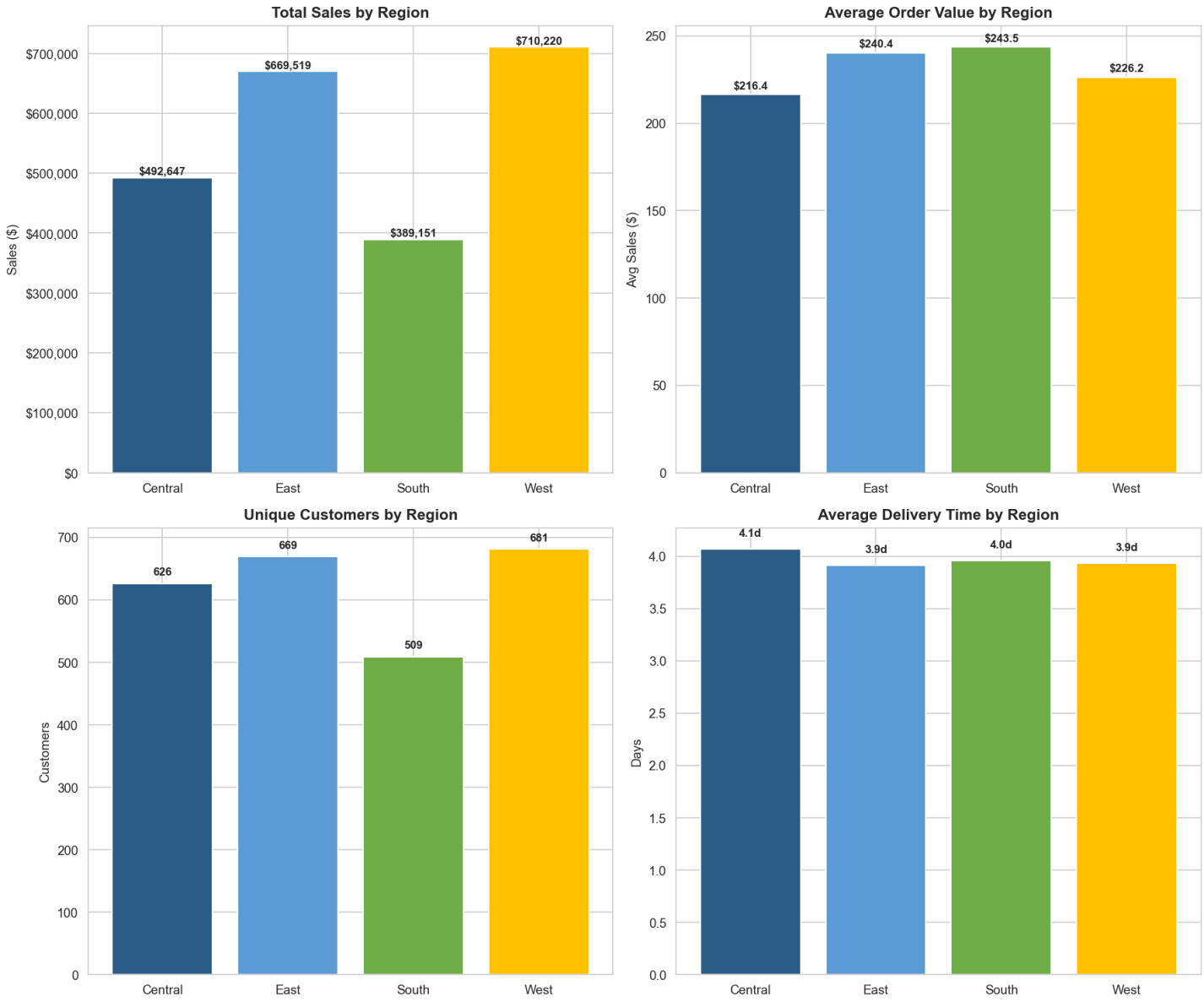


Figure A3: Regional performance dashboard showing total sales, average order value, customer counts, and delivery times.

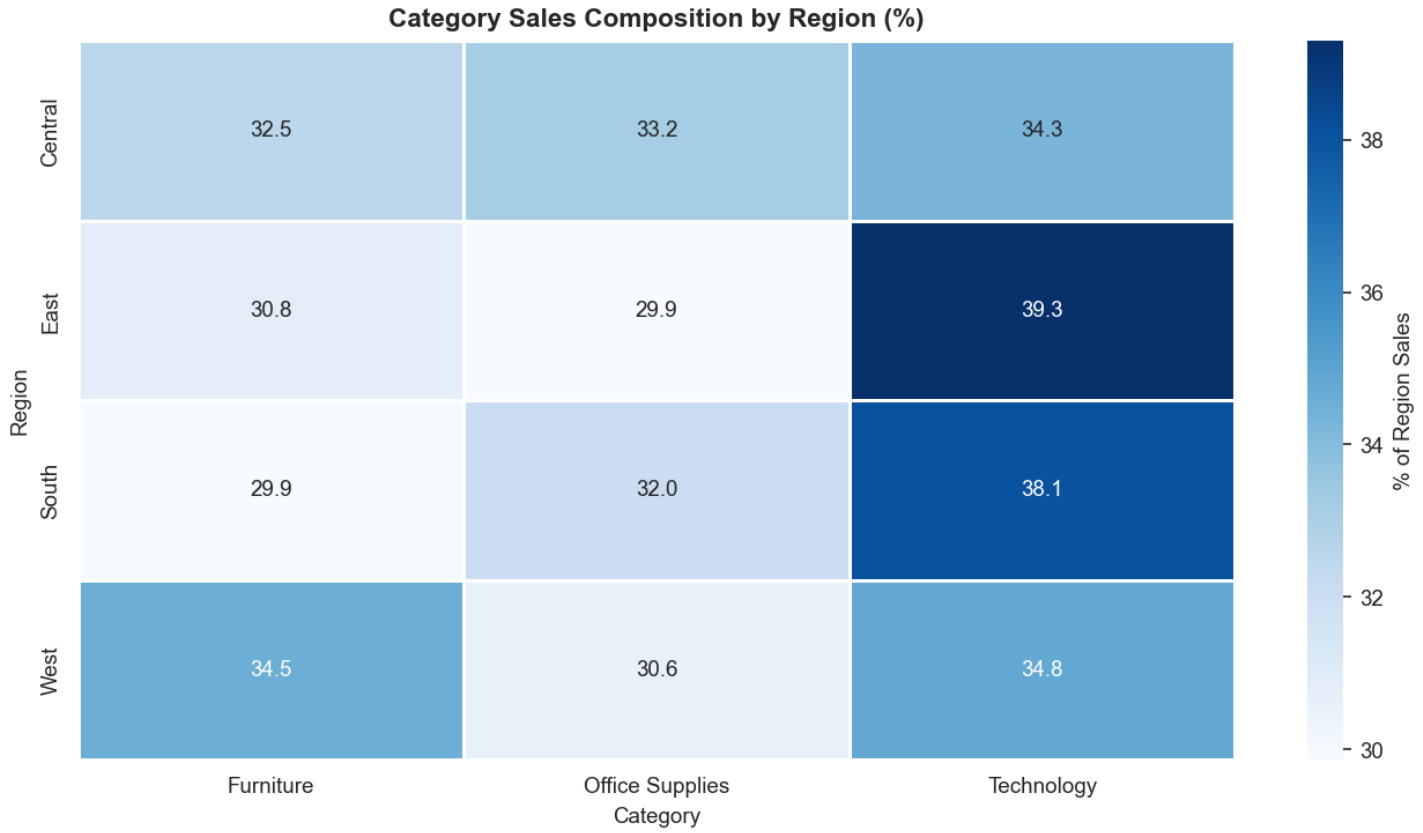


Figure A4: Category composition by region (%). Technology dominates in the West; Office Supplies in the South.

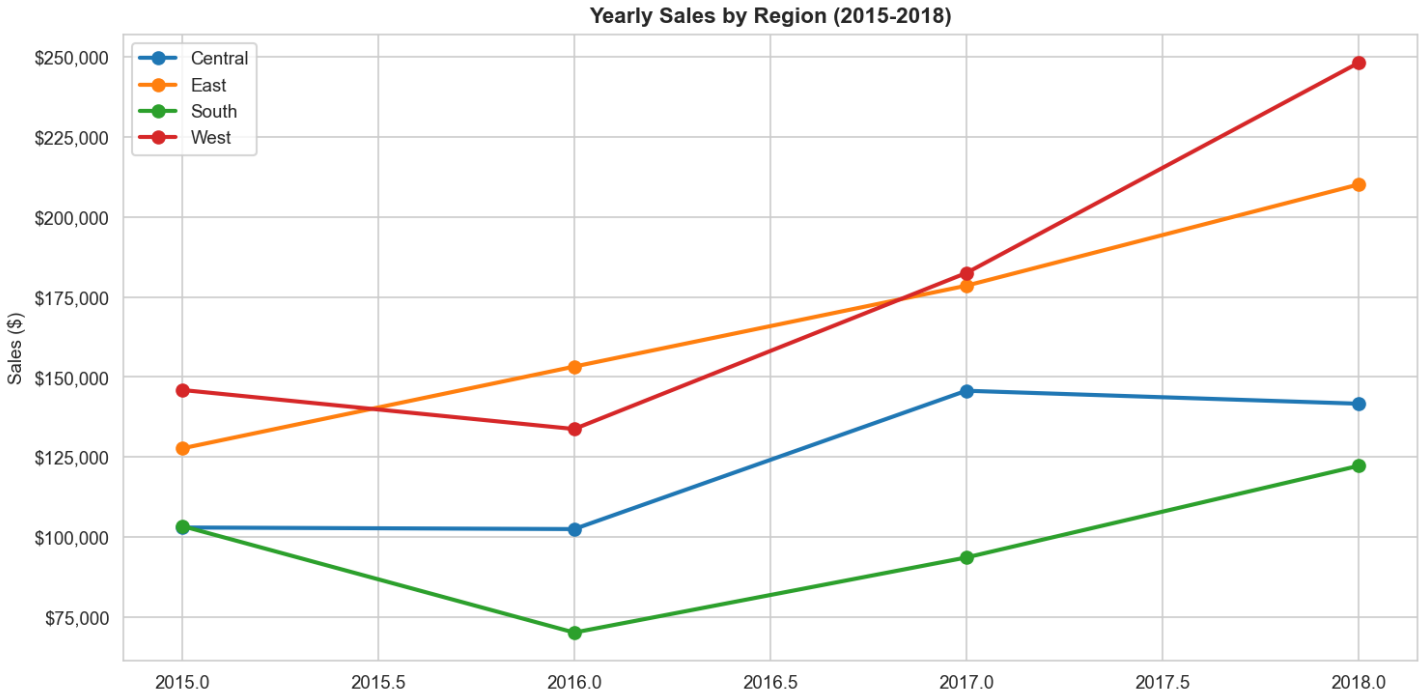


Figure A5: Yearly sales trends by region. All regions growing, with West accelerating post-2017.

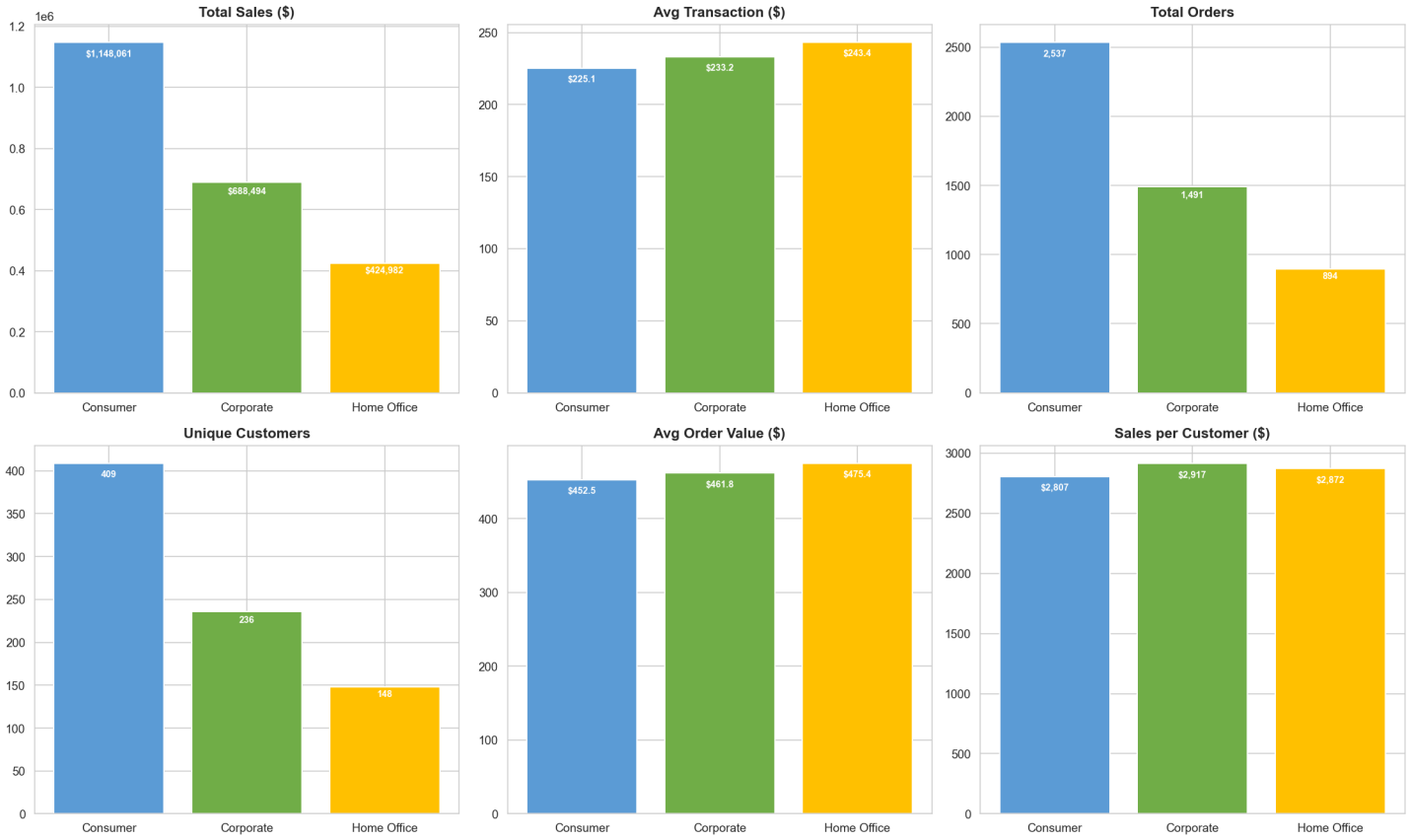


Figure A6: Segment value dashboard showing total sales, average transaction, order counts, and per-customer metrics.

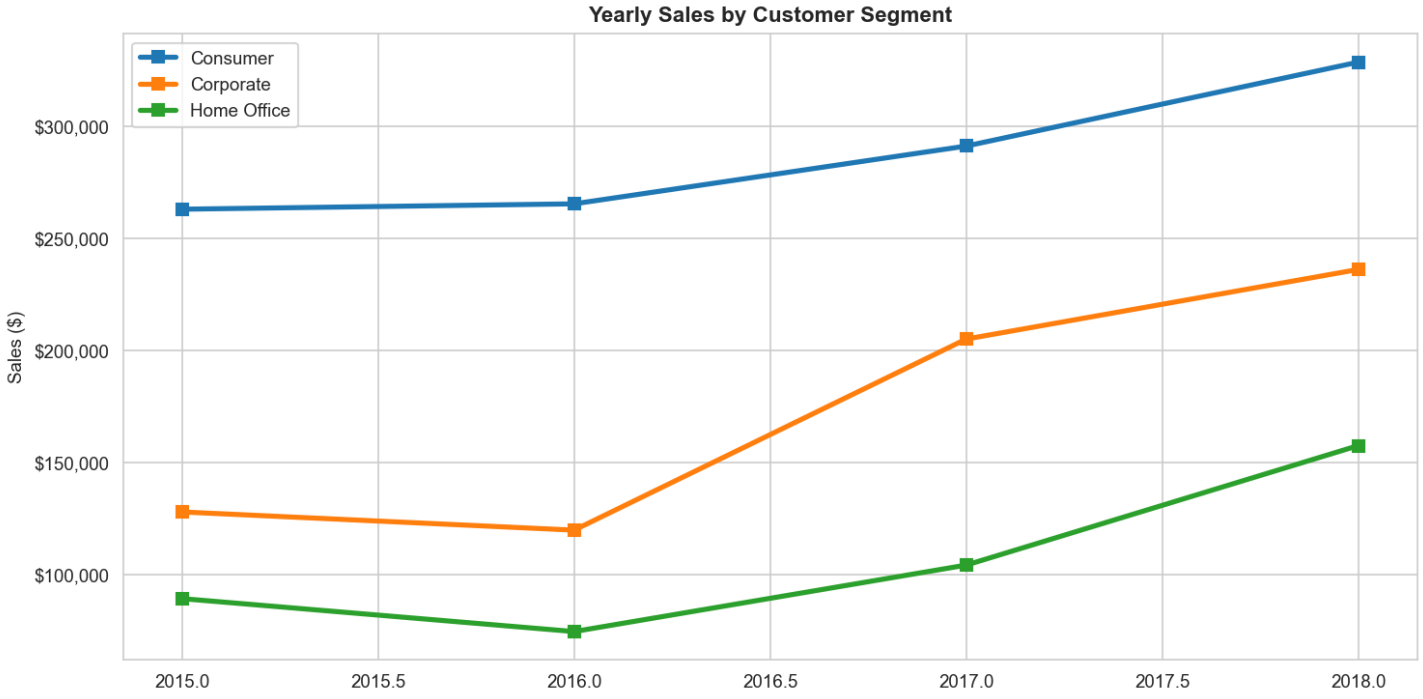


Figure A7: Yearly sales by customer segment. All three segments showing consistent growth.

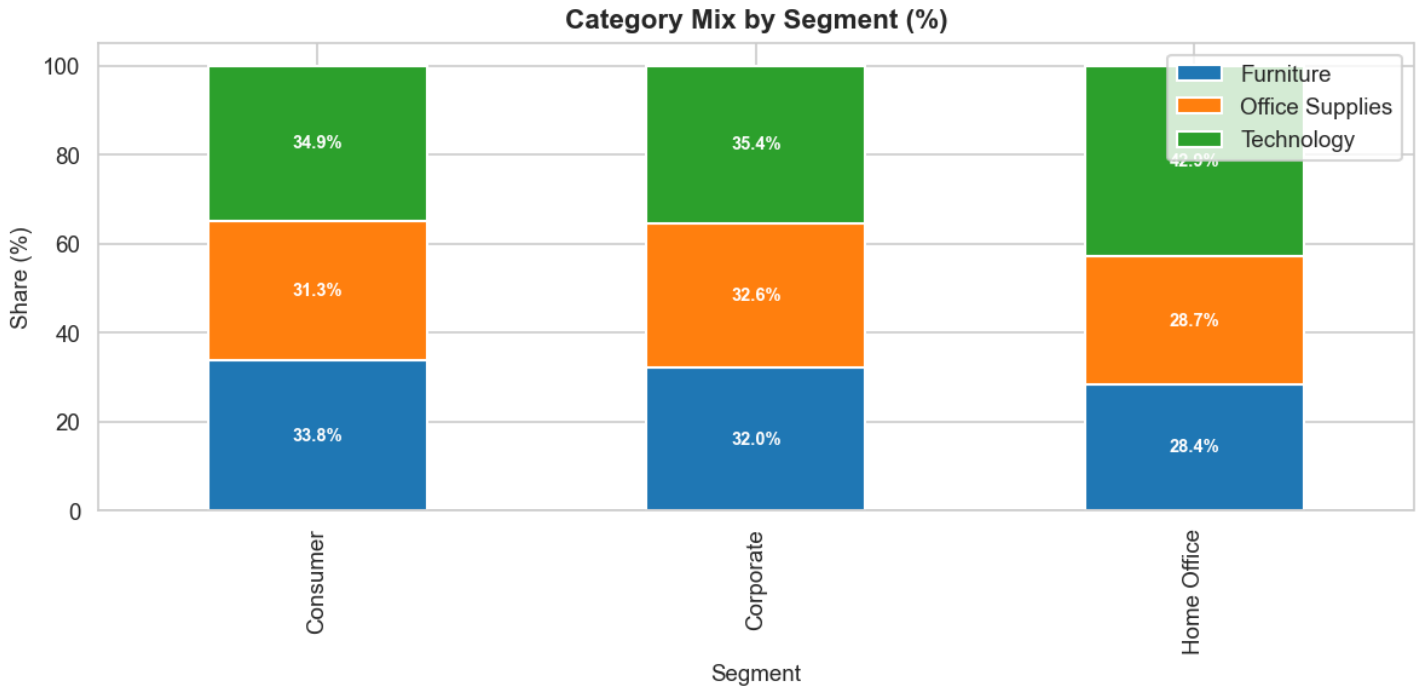


Figure A8: Category mix by customer segment (%). Corporate customers favour Technology.

10. Methodological Note

This analysis was conducted using a structured four-stage framework: Description, Inquiry, Goal Design, and Analysis/Visualisation. This approach ensures rigour, transparency, and traceability throughout the analytical process.

All data processing and visualisation was performed in Python 3.12 using the following libraries: pandas for data manipulation, matplotlib and seaborn for visualisation, and fpdf2 for report generation. The full analysis script and all output files (PNG figures, CSV tables) are available alongside this report.

Data quality was assessed prior to analysis and found to be excellent (99.8% complete). The 11 missing Postal Code entries were identified but did not affect the analyses conducted, as Region-level and Year-level aggregates were used. Sales outliers above the 99th percentile ($2,480) were retained but noted for sensitivity considerations.

The analytical framework supports structured reasoning and reproducibility. Each analysis track includes clear data subset definitions, metric calculations, visualisation choices, and interpretative narratives to ensure findings are both evidence-based and actionable.